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Unit 2 - Bringing Ideas to Life

PDF Documents

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

INTRODUCTION

In this unit, you will take your understanding of BMI and begin designing a device that can calculate and display a user's BMI. To create this device, you'll need to think carefully about the inputs, calculations, and outputs that will be required. Let's dive into the process of turning your idea into a functional device.

IDEA GENERATION

Your BMI-measuring device will need to allow users to input their measurements, process this data, and then display the calculated BMI. Think about the following aspects:

1. USER INPUT

Consider how a user might easily input their weight and height. What type of mechanism could they use to select or adjust these values?

2. PROCESSING THE DATA

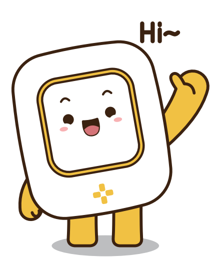
To calculate BMI, the device will need to process the inputted values. What could assist in performing this calculation automatically?

3. DISPLAYING RESULTS

After calculating the BMI, the user needs to see their result clearly. What might be the best way to show this information?

TIPS

You might want to explore options that allow the user to adjust values with ease and see their BMI displayed immediately. Think about tools that make it easy to input and visualize numerical data.

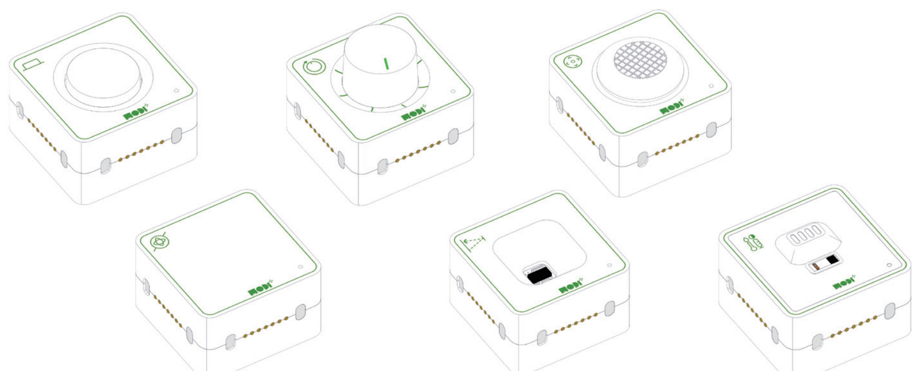


Required Modules & Brainstorming

Now that you have some ideas about how your device should work, let's think about the specific components you'll need:

Input Mechanism

How can you allow the user to input or select their weight and height? Imagine a component that can be turned or pressed to adjust these values.

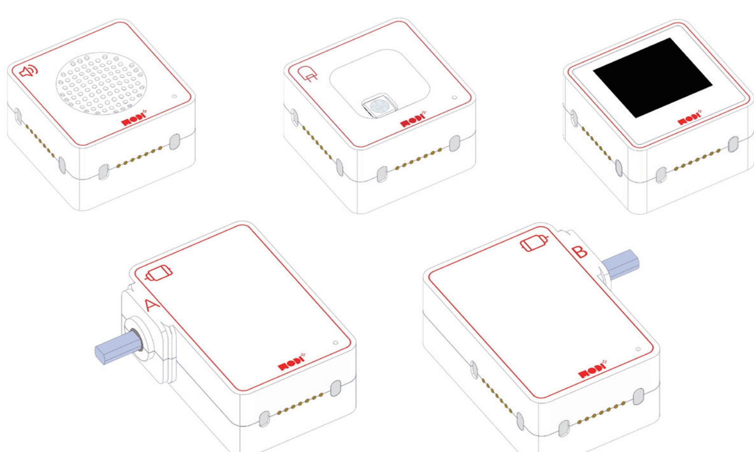


Calculation Process

You'll need a way to process the input data and calculate the BMI automatically. Consider a tool that could help manage and adjust the inputted data for accurate calculations.

Output Display

How will the result be displayed to the user? Think of a way to present numerical information in a clear and understandable manner.



Required Modules & Brainstorming

Which MODI modules do you need for the creation?

DIAL

BUTTON

ENVIRONMENT

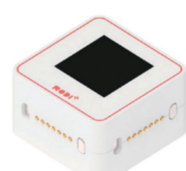
JOYSTICK

TOF

SPEAKER

IMU

LED

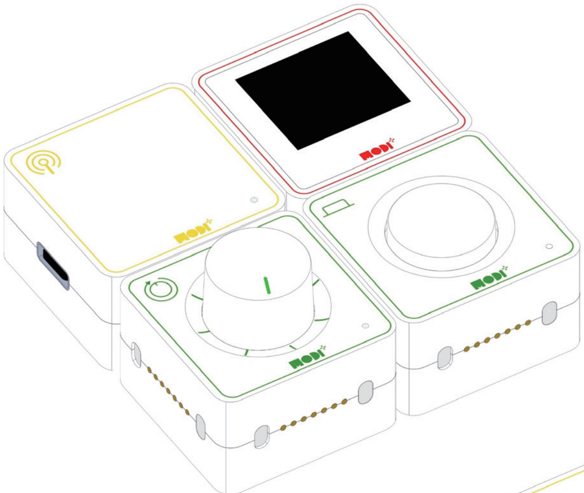


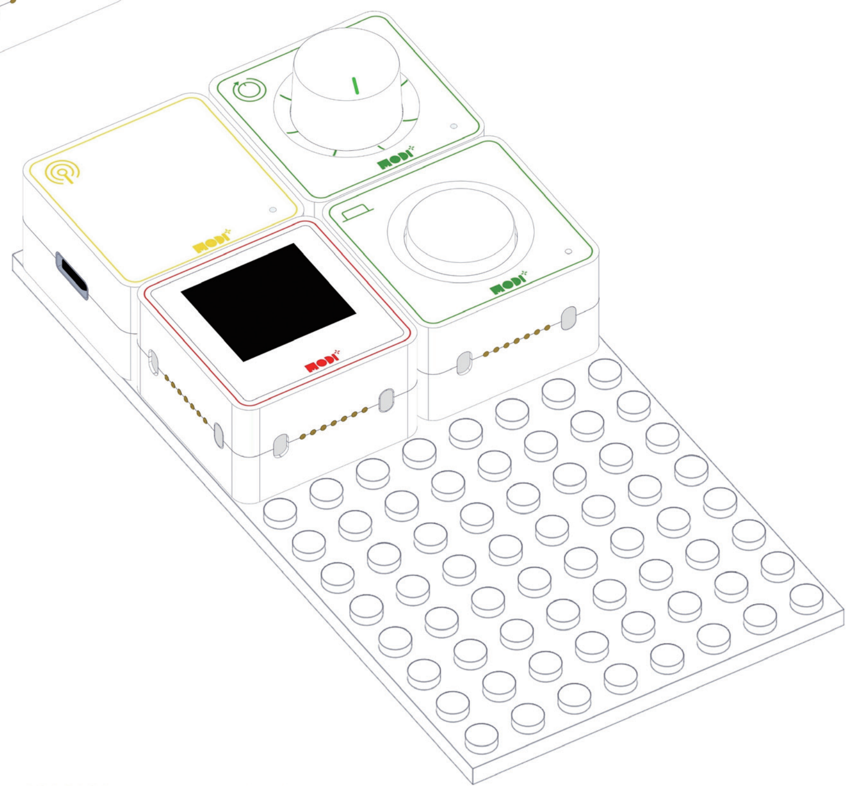
DISPLAY

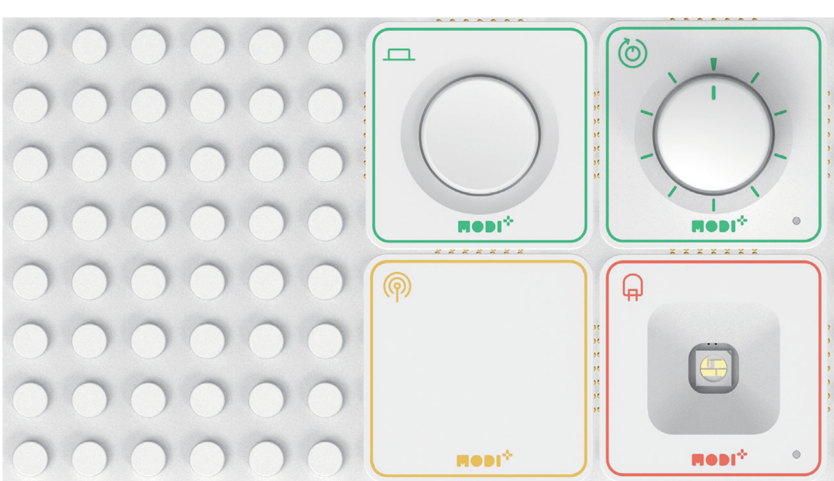
MOTOR A

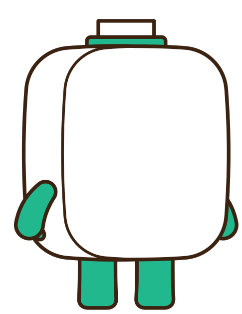
MOTOR B

WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.



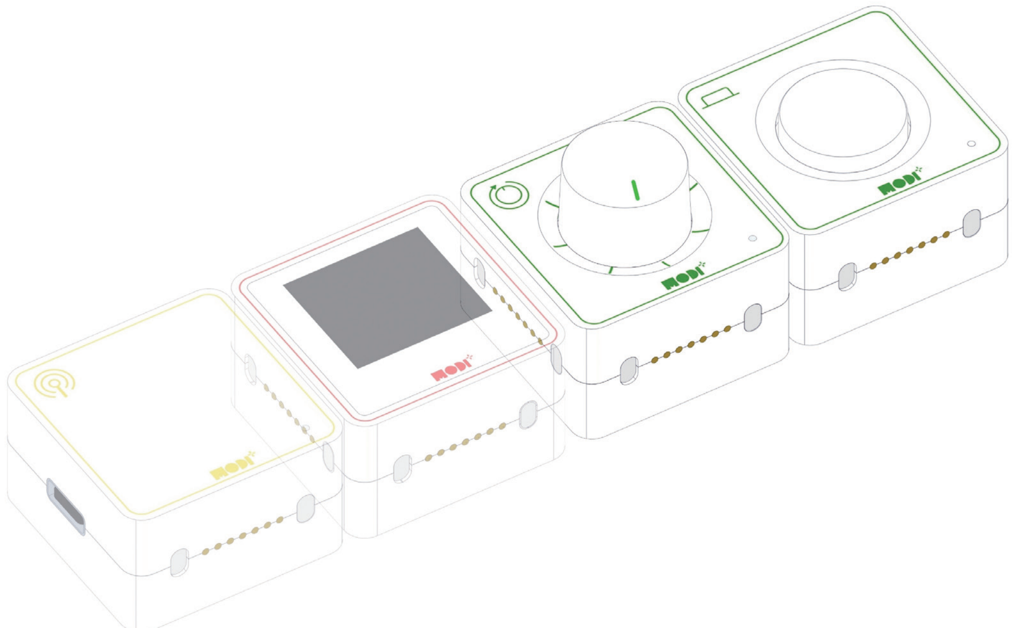






Module Functionality

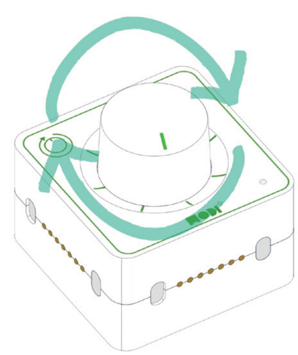
In your BMI-measuring device, the Dial and Button modules are essential components for user interaction, allowing users to input and control the device's functions effectively. Here's how each of these modules contributes to the functionality of your device:



APPLICATIONS IN FLOOD ALARM SYSTEM

HOW IT WORKS:

The Dial module acts as a rotary input device. It allows the user to adjust numerical values by turning the dial. For the BMI-measuring device, the Dial will be used to input the user's height and weight. When the user rotates the dial, it changes the values on the display module, allowing them to select their specific measurements accurately.



ROLE IN THE DEVICE:

- **Adjusting Values:** The Dial module lets the user scroll through different values of height and weight, providing a precise way to input these key measurements.
- **Interactive Input:** It offers an intuitive way to interact with the device, making it easy for users to set their measurements by simply turning the dial.

BUTTON MODULE

HOW IT WORKS:

The Button module is a simple input device that detects when it is pressed. In the BMI-measuring device, the Button is used to confirm the values entered by the Dial. Once the user has adjusted the height and weight to the correct values, pressing the Button will lock in these numbers and trigger the calculation of the BMI.

ROLE IN THE DEVICE:

- **Confirmation:** The Button is used to confirm the user's input after they have set their height and weight using the Dial.
- **Triggering the Calculation:** Once the values are confirmed by pressing the Button, the device will proceed to calculate the BMI based on the input values.

EXAMPLE IN ACTION:

- **Confirming Height and Weight:** After adjusting the height and weight with the Dial, the user presses the Button to confirm these values. This action finalizes the input and allows the device to calculate and display the BMI.